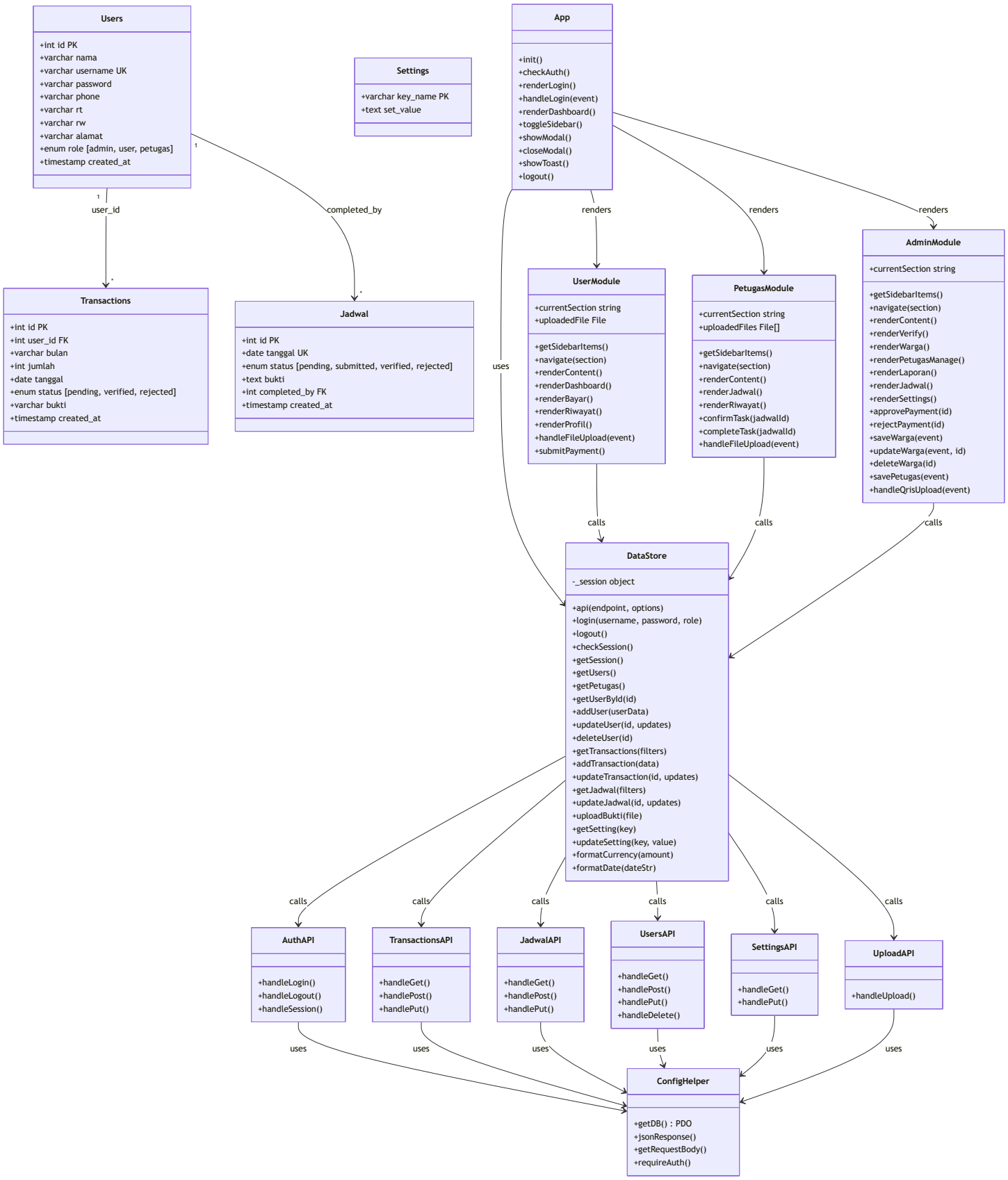
🏗️ Class Diagram — SIPARES

Sistem Pembayaran Retribusi Sampah

Diagram


Users
+int id PK
+varchar nama
+varchar username UK
+varchar password
+varchar phone
+varchar rt
+varchar rw
+varchar alamat
+enum role [admin, user, petugas]
+timestamp created_at
Settings
+varchar key_name PK
+text set_value
App
+init()
+checkAuth()
+renderLogin()
+handleLogin(event)
+renderDashboard()
+toggleSidebar()
+showModal()
+closeModal()
+showToast()
+logout()
1
1
user_id
completed_by
*
*
Transactions
+int id PK
+int user_id FK
+varchar bulan
+int jumlah
+date tanggal
+enum status [pending, verified, rejected]
+varchar bukti
+timestamp created_at
Jadwal
+int id PK
+date tanggal UK
+enum status [pending, submitted, verified, rejected]
+text bukti
+int completed_by FK
+timestamp created_at
uses
renders
renders
renders
UserModule
+currentSection string
+uploadedFile File
+getSidebarItems()
+navigate(section)
+renderContent()
+renderDashboard()
+renderBayar()
+renderRiwayat()
+renderProfil()
+handleFileUpload(event)
+submitPayment()
PetugasModule
+currentSection string
+uploadedFiles File[]
+getSidebarItems()
+navigate(section)
+renderContent()
+renderJadwal()
+renderRiwayat()
+confirmTask(jadwalId)
+completeTask(jadwalId)
+handleFileUpload(event)
AdminModule
+currentSection string
+getSidebarItems()
+navigate(section)
+renderContent()
+renderVerify()
+renderWarga()
+renderPetugasManage()
+renderLaporan()
+renderJadwal()
+renderSettings()
+approvePayment(id)
+rejectPayment(id)
+saveWarga(event)
+updateWarga(event, id)
+deleteWarga(id)
+savePetugas(event)
+handleQrisUpload(event)
calls
calls
calls
DataStore
-_session object
+api(endpoint, options)
+login(username, password, role)
+logout()
+checkSession()
+getSession()
+getUsers()
+getPetugas()
+getUserById(id)
+addUser(userData)
+updateUser(id, updates)
+deleteUser(id)
+getTransactions(filters)
+addTransaction(data)
+updateTransaction(id, updates)
+getJadwal(filters)
+updateJadwal(id, updates)
+uploadBukti(file)
+getSetting(key)
+updateSetting(key, value)
+formatCurrency(amount)
+formatDate(dateStr)
calls
calls
calls
calls
calls
calls
AuthAPI
+handleLogin()
+handleLogout()
+handleSession()
TransactionsAPI
+handleGet()
+handlePost()
+handlePut()
JadwalAPI
+handleGet()
+handlePost()
+handlePut()
UsersAPI
+handleGet()
+handlePost()
+handlePut()
+handleDelete()
SettingsAPI
+handleGet()
+handlePut()
UploadAPI
+handleUpload()
uses
uses
uses
uses
uses
uses
ConfigHelper
+getDB() : PDO
+jsonResponse()
+getRequestBody()
+requireAuth()

Penjelasan Relasi

Relasi Database (Entity)

Relasi	Tipe	Keterangan
Users → Transactions	One-to-Many	Satu user (warga) bisa memiliki banyak transaksi. Foreign Key: `user_id`
Users → Jadwal	One-to-Many	Satu user (petugas) bisa menyelesaikan banyak jadwal. Foreign Key: `completed_by`

Relasi Backend (API Layer)

Dari	Ke	Tipe	Keterangan
Semua API	ConfigHelper	Dependency	Semua API menggunakan helper untuk koneksi DB, auth, dan response

Relasi Frontend

Dari	Ke	Tipe	Keterangan
App	DataStore	Dependency	App menggunakan DataStore untuk komunikasi API
App	UserModule / PetugasModule / AdminModule	Composition	App merender modul sesuai role user
UserModule / PetugasModule / AdminModule	DataStore	Dependency	Setiap modul memanggil DataStore untuk operasi data
DataStore	Semua API	Association	DataStore memanggil endpoint API via fetch()